

COLLINS BAY, SK, Canada

WMO: 710750

Lat: 58.229N

Lon: 103.678W

Elev: 1354

StdP: 13.99

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.1	-31.3	-42.4	0.5	-35.1	-38.2	0.6	-31.1	18.9	-0.5	17.2	-2.0	5.3	290	0.745

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.0	78.5	59.6	75.1	58.3	71.8	57.3	62.7	72.6	61.0	70.4	59.4	68.1	8.8	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.4	79.4	65.1	57.6	74.4	63.5	55.7	69.4	62.5	29.1	72.6	27.9	70.3	26.7	68.4	70.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.7	15.7	14.1	-40.2	84.8	5.2	3.2	-44.0	87.2	-47.0	89.0	-49.9	90.8	-53.7	93.2		
(5)			-40.5	66.1	5.2	2.0	-44.2	67.5	-47.3	68.6	-50.2	69.7	-54.0	71.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.6	-9.1	-4.2	8.1	25.2	39.9	54.5	60.3	57.3	46.4	31.1	11.1	-3.3		
	DBStd	26.96	14.61	13.36	14.96	13.36	10.63	8.11	6.24	6.30	8.43	9.09	12.55	14.66		
	HDD50	9369	1832	1518	1299	746	345	46	3	11	164	587	1166	1652		
	HDD65	14058	2297	1938	1764	1195	778	326	170	249	558	1050	1616	2117		
	CDD50	836	0	0	0	1	33	180	324	238	58	2	0	0		
	CDD65	49	0	0	0	0	1	10	26	11	1	0	0	0		
	CDH74	400	0	0	0	0	14	85	199	91	11	0	0	0		
	CDH80	53	0	0	0	0	3	8	28	13	1	0	0	0		
	WSAvg	7.2	6.5	7.2	7.6	7.7	7.7	7.2	6.8	6.9	7.8	7.7	6.7	6.4		
	PrecAvg	21.0	0.8	0.7	1.1	1.0	1.6	2.1	3.2	3.5	2.6	1.9	1.2	0.7		
(15) Precipitation	PrecMax	29.4	1.7	1.9	3.1	3.1	5.6	5.0	6.3	6.9	6.5	3.5	2.5	1.5		
	PrecMin	14.3	0.0	0.1	0.1	0.1	0.3	0.3	1.1	0.7	0.9	0.8	0.3	0.0		
	PrecStd	4.1	0.5	0.5	0.7	0.8	1.3	1.1	1.2	1.6	1.4	0.7	0.6	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.3	32.2	46.8	59.6	74.7	80.8	83.9	81.9	74.6	58.3	40.7	30.9		
		MCWB	30.3	30.4	37.4	45.0	54.2	59.4	62.0	61.5	58.7	47.9	36.1	30.2		
	2%	DB	24.6	26.6	40.0	52.2	68.0	76.3	78.9	76.4	67.5	51.5	34.0	25.1		
		MCWB	24.1	25.5	33.3	40.9	50.7	57.2	60.5	60.0	55.3	44.9	32.2	24.4		
	5%	DB	18.7	21.3	34.9	47.7	62.5	72.5	75.7	72.2	62.9	46.5	31.1	21.0		
		MCWB	18.1	20.3	30.8	38.9	48.3	55.7	59.2	58.3	54.0	42.2	29.9	20.5		
	10%	DB	12.1	15.9	30.0	43.2	57.2	68.7	72.4	68.5	58.6	42.8	28.1	16.9		
		MCWB	11.6	14.9	27.2	35.9	46.1	54.3	58.3	57.1	51.5	40.4	27.2	16.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.6	30.6	37.7	46.0	56.5	63.8	66.2	64.3	60.6	50.6	36.7	29.9		
		MCDB	31.0	32.0	45.7	57.0	70.3	72.9	75.3	77.8	69.9	56.6	40.5	30.9		
	2%	WB	24.4	25.4	33.9	41.9	52.4	60.5	63.4	61.9	57.4	45.6	32.6	24.8		
		MCDB	24.8	26.5	39.0	51.2	64.8	69.7	72.6	72.1	65.2	49.5	33.7	25.3		
	5%	WB	18.1	20.3	31.3	39.3	49.6	58.0	61.6	60.1	55.0	42.9	29.8	20.6		
		MCDB	18.5	21.3	35.1	46.8	60.3	68.1	70.7	69.2	60.9	45.7	30.9	21.2		
	10%	WB	11.2	15.2	27.2	36.5	46.6	55.9	60.0	58.5	52.7	40.4	27.1	16.5		
		MCDB	11.7	16.1	30.0	42.6	56.4	66.2	68.8	65.9	58.0	43.1	27.9	17.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.5	16.2	19.5	19.1	18.5	17.7	17.0	15.6	13.8	9.6	11.2	12.4		
		MCDBR	16.6	17.8	20.7	21.1	26.3	22.9	22.0	21.3	20.5	15.1	11.3	14.3		
		MCWBR	16.9	16.5	15.4	12.5	13.8	9.8	8.8	9.4	11.1	9.7	9.2	14.5		
		MCDBR	16.7	17.3	19.9	19.9	24.4	19.1	18.2	18.2	18.3	13.3	10.7	14.5		
	5% WB	MCWBR	16.5	16.4	15.1	12.3	13.5	9.6	8.7	9.3	10.8	9.1	9.4	14.3		
(40) Clear-Sky Solar Irradiance	taub		0.211	0.260	0.258	0.312	0.328	0.354	0.392	0.362	0.303	0.278	0.254	0.191		
	taud		2.213	2.198	2.301	2.273	2.446	2.438	2.346	2.446	2.605	2.537	2.145	2.120		
	Ebn at Noon		216	248	287	285	288	280	265	266	272	247	181	183		
	Edh at Noon		14	24	29	35	32	33	36	30	22	17	15	11		
(44) All-Sky Solar Radiation	RadAvg		199	523	1074	1617	1827	1845	1674	1323	860	402	168	117		
	RadStd		17	42	63	127	171	192	97	105	105	45	21	14		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page